

COGNITIVE SCIENCE

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knowledge: debate

← Plato →
John Locke → blank slate - knowledge comes from experience
Leibniz → there has to be some structure that isn't learned

EVOLUTIONARY LEARNING

→ We have lots of knowledge that isn't learned

SOCIAL LEARNING + CULTURE

→ argues that lots of interesting human behaviours
are product of
culture + evolution

Natural selection → environmentally filtering - adaptation

behavioural traits / physical traits

eg ← beavers building dams
beats - mammals grooming
blood

Needs to be ← heritable
variations in the population
an advantage for those who
have it

→ can it help understand evolution of cognitive traits?

(2)

Natural selection vs Social learning

↓

favored by

↓

Evolutionary

(EP)

Psychology

founders

4 claims

human brain

1) product of natural selection

2) adapted to tackle problems

3) now is our ancestors but evolved

3) cognitive capacities — heritable

How Should we understand the role of natural selection

Our skulls house Stone age minds (Cosmides / Tooby)

10000 years ago

small hunter groups
not environment} (EP) says our brain
evolved to solve
this

Industrial Rev. 200 y.a.

Agricultural 5000 y.a.

Some cognitive capacities

now

Maladaptation

Now

← were meant for
beforevisceral reaction to Snake/spiders
(Stephen Pinker)(EP) craving
salt
fat
sugar

issues before — maintain harmonious small group

EP says → {
our brain
evolve to solve {
cooperation
who to share with
is someone taking advantage

consider, Tooby
we are good at ← social norms
detecting, → being violated
but bad at generalizing this
to logically equivalent situations

Warren selection task
(Cheater detection problem)
(Peter Warren)

Cards example

EP view → brain as Swiss army knife → many mini brains
specific for each task

ER → brain as chef knife → 1
good blade that allows us
to do everything (even if not
tailored for each task)

many ≠ small adaptation vs. one big adaptation / change

eg. how the eye changed incrementally

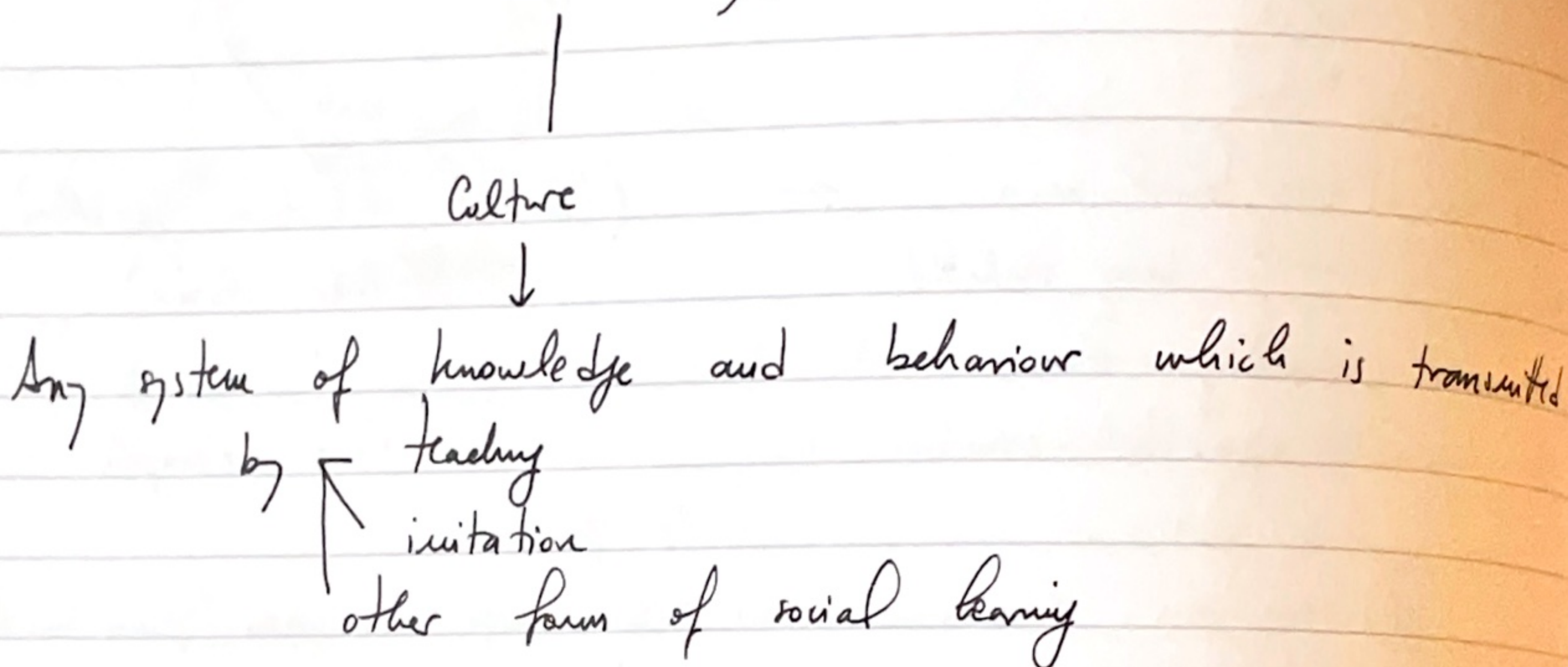
parts of the brain
can work independently

modularity of human mind

③

- Ecological inheritance

- Social learning → important non-genetic mechanism for transmission of knowledge and behaviours in human populations —



Cumulative culture → one generation builds on knowledge passed on from previous generation.

↓
(vs. biological adaptation
e.g. beaver)

↓
lactone persistence

most likely = cultural + biological evolution

The expressive power of language

because it has rules

it is FLEXIBLE

is it an adaptation? (Pinker, Bloom) → communicative power

are some of the structural properties a product of culture (over biology)?

people modify the language to meet their communicative needs ←

(EP) → game view of evolution → an organism responding to challenges imposed by the environment

Niche constructing → how organisms alter their environment
|
setting selective pressures